Coasters and pendulums

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use height differences to predict speed in ideal motion.

SUCCESS CRITERIA - I can...

1. Identify the relevant vertical height change.
2. Conserve energy under stated assumptions.
3. Separate path length from gravitational energy change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pendulum” and “potential energy” to compare your model or result. Sentence starters are optional.

pendulum: an object that swings about a fixed support

potential energy: energy associated with the configuration of interacting objects

kinetic energy: energy associated with motion

01 Bring back an earlier idea

Gravitational potential energy changes from -15 J to -3 J.
Find ΔU and state whether energy is gained or lost.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $mg\Delta h = \frac{1}{2}mv^2$; $v = \sqrt{(2\ g\Delta h)}$.
Point mass released from rest, no losses or rotation; Δh is the positive vertical drop.

Worked reasoning: A pendulum released from rest 0.8 m above its lowest point reaches speed $\sqrt{(2\ gh)}=4$ m/s there with g=10 and negligible losses.

Your turn: Sketch the path and energy bars at the release, bottom and opposite turning point.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Two frictionless tracks start and end at the same heights but take different paths.
Compare final speeds from rest.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart drops 3 m.
Use g=10.
Find lost potential energy and gained kinetic energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Coasters and pendulums

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A classmate uses total ramp length as h in mgh.
Explain how to identify the correct measurement.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A fixed impulse changes a target’s sideways velocity by 0.04 m/s.
Find the modeled shift after 600 s and state a limitation.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

A pendulum loses 12 J of gravitational energy while moving down. Neglect losses. What is its kinetic energy gain?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A pendulum returns to only 0.6 m above the bottom. For a 1 kg bob with g=10, find energy transferred out of its mechanical-energy account.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3